

EDUCATION

Worcester Polytechnic Institute

B. S. Robotics Engineering; B. S. Computer Science

○ Relevant Classes: *Deep Learning For Perception, Swarm Intelligence, AI for Robotics, Unified Robotics: Manipulation, Navigation*

May 2025

Worcester, MA

WORK EXPERIENCE

Smartapp.com – R&D Division

Autonomy Engineering Intern

Expanded on a Python and React-TS thrust test stand hosted on a Raspberry Pi, with customizable datalogging and test procedures

Developed an intuitive and flexible motor library, which enables ‘hot-swapping’ of intelligent motor classes and objects

Optimized proprietary robot locomotion techniques in Nvidia’s IsaacSim using deep reinforcement learning (PyTorch and PPO)

May 2024 – Aug 2024

Worcester, MA

OpenSTEM: Experiential Robotics Platform (XRP)

Lead Software Developer, XRPLib

Designed and maintained a modular MicroPython-based robotics control library deployed across 20K+ classroom-scale robots

Managed a team of 1-5 over 2.5 years while coordinating with corporate partners and engineers from DEKA and Sparkfun

Aug 2022 – Present

Worcester, MA

Worcester Polytechnic Institute

Undergraduate Student Assistant

Taught algorithm design (Dijkstra) and code debugging skills while leading lab sections for a new course (RBE 100x) using XRP robots

Oct 2022 - Dec 2022

Worcester, MA

Jacobs Technology – Jacobs Software Engineering Center

Software Engineering Intern

Refactored and extended C#/.NET mission support tooling to manage dynamic SQL filter sets for DAFIF airspace pathing datasets

May 2022 – Aug 2022

Hudson, NH

PROJECTS

Terrawarden Drone Cleanup – Major Qualifying Project

Developed aerial manipulator for autonomous waste collection using Intel RealSense and YOLOv11 with low-latency inference (<3ms)

Implemented object detection model trained on procedurally rendered Blender dataset, enabling robust generalization to roadside litter

Architected ROS2-based state machine to isolate high-level behavior, programmed using Test-Driven Development (PyTest)

Aug 2024 – May 2025

Drone Racing Gate Semantic Segmentation

Used Blender to generate a dataset of 5000 images of drone racing gates for training a semantic segmentation model (U-Net)

Applied various image augmentation techniques, such as gaussian blur and color jitter, to increase the robustness of the trained model

Aug 2024 – Oct 2024

Video Game AIs

Benchmarked the abilities of an algorithmic model (Expectiminimax) vs a reinforcement model (Q-Learning) at playing Bomberman

Implemented imitation learning on a Deep Q-Learning model in Pytorch, training it to play the Snake Game

Jan 2024 – Mar 2025

Robotic Navigation – SLAM and AMCL

Developed a robot that could autonomously navigate and map an unknown space, then localize itself when relocated at a later time

Implemented Simultaneous Localization and Mapping (SLAM), AMCL, A*, and Pure Pursuit on a Turtlebot3 with a planar LiDAR

Oct 2023 – Dec 2023

Hospital Wayfinding & Facilities System

Developed JavaFX-based kiosk application with Dijkstra-based pathfinding, signage customization, and service requests over SQL backend

Led 11-member Agile team in design, implementation, and user testing phases with modular MVC and OOP design patterns.

Mar 2023 – May 2023

TECHNICAL SKILLS

Perception & Sensing	Intel Realsense, LiDAR, Image Segmentation, Object Detection, Kalman Filters, YOLOv11
ML / RL Tools	PyTorch, Scikit-Learn, Nvidia IsaacLab, DQN, Imitation Learning, ResNet, Pandas, W&B
Motion Planning	ORCA, SLAM, (gmapping), AMCL, A*, Pure Pursuit, Monocular Depth Estimation
Simulation	Nvidia IsaacSim, Gazebo, Unity Game Engine, MuJoCo, Blender (Synthetic Data Generation), PyGame
Embedded Systems	MicroPython, Arduino, Raspberry Pi, Nvidia Jetson, UART/I ² C, ROS/ROS2, HALs
Version Control & Dev Tools	Git, Agile, Kanban, Azure DevOps, Jira, Figma, Github Projects, Microsoft Office
Languages	Python, TypeScript, C++, C#, Java, C, JavaScript, Kotlin, MATLAB

EXTRACURRICULARS

WPI Robotics Prototyping Club, Founder, Treasurer

WPI Cooking Club, President

WPI Rho Beta Epsilon, Alpha Chapter

WPI Bowling Club, Treasurer

WPI VexU, Software Co-Lead

Scouting America, Troop 106, Eagle Scout

Aug 2024 - Present

Apr 2023 - Present

Feb 2025 - Present

Aug 2022 – Feb 2023

Aug 2022 – Feb 2023

Mar 2014 – July 2021